EXCELLENCE

CLASS 11 - 2026-27

Physics | Vector 05

Vectors

Vectors in Physics

By - Excellence Programme

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Topics to be covered

01 Unit vector and direction practice

02 Forces and resultants in plane

03 Equal vectors at 60°

04 Coulomb and gravity force vectors

05 True / false on zero resultant

06 Change in velocity

07 Position vector and displacement

08 Dot product quick practice

What we will do

Use unit vectors to fix magnitude and direction separately.

Add force vectors from diagrams and from components.

Treat Delta v and displacement as vector subtraction problems.

Keep distance (scalar path) separate from displacement (vector).

Speed can stay the same while velocity changes a lot.

Same magnitude, chosen direction

Vector = magnitude × unit vector in the required direction

If C has magnitude of A and direction of B:

$$\mathbf{C = |A|\,(B / |B|)}$$

Do not write C = A. Magnitudes match, but the arrow of C follows B.

Find β for a unit vector

$A = 0.4\,\hat{i} + 0.3\,\hat{j} + \beta\,\hat{k}$ is a unit vector. Find β.

Solution: Find β for a unit vector

Step 1: $|A| = 1$, so $\sqrt{(0.16 + 0.09 + \beta^2)} = 1$

Step 2: $0.25 + \beta^2 = 1$, so $\beta^2 = 0.75$

$\beta = \pm \sqrt{(3)}/2$.

Rough work / extra work

Same |A|, direction of B

$A = 2\hat{i} + \hat{j} + 2\hat{k}$ and $B = \hat{i} + \hat{j} + 2\hat{k}$. Find C with $|C| = |A|$ and direction along B.

Solution: Same |A|, direction of B

Step 1: $|A| = \sqrt{(4 + 1 + 4)} = 3$

Step 2: $|B| = \sqrt{(1 + 1 + 4)} = \sqrt{(6)}$

Step 3: $C = 3\,(B / |B|)$

$C = 3\,(\hat{i} + \hat{j} + 2\,\hat{k}) / \sqrt{(6)}.$

Force parallel to a given vector

A force of magnitude 20 N acts along $\vec{A} = 2\hat{i} + 3\hat{j} - \sqrt{3}\hat{k}$. Mass is 5 kg. Find F and a.

Solution: Force parallel to a given vector

Step 1: $|A| = \sqrt{(4 + 9 + 3)} = 4$

Step 2: $F = 20\ (A / 4) = 5\ (2\ \hat{i} + 3\ \hat{j} - \sqrt{3}\ \hat{k})$

Step 3: $a = F / m = 2\ \hat{i} + 3\ \hat{j} - \sqrt{3}\ \hat{k}$

$F = 5(2\ \hat{i} + 3\ \hat{j} - \sqrt{3}\ \hat{k})$, $a = 2\ \hat{i} + 3\ \hat{j} - \sqrt{3}\ \hat{k}$.

Rough work / extra work

Net force decides the direction of motion

For a point mass at rest, acceleration is along F_net.

$$F_{net} = F_1 + F_2 + F3$$

Cancel like components first. Then read the leftover direction.

If F_net = 0, the object stays at rest. Do not invent a direction.

Three forces on a point object

Forces 2 î + 2 ĵ, -2 î - 2 ĵ and -4 î act on a point object at rest. The object moves along:

1. x-axis
2. y-axis
3. z-axis
4. Object does not move

Solution: Three forces on a point object

Step 1: $F_{net} = (2 - 2 - 4)\,\hat{i} + (2 - 2)\,\hat{j} = -4\,\hat{i}$

Option 1: along the negative x-axis.

Extra force so that R is only along y

Forces: 1 N at 60° above +x, and 2 N at 30° left of +y. Minimum additional force so that F_net is only along y?

1. √3 N
2. √3/2 N
3. 1.5 N
4. 1/2 N

Solution: Extra force so that R is only along y

Step 1: x from 1 N: + cos 60° = +1/2 N

Step 2: x from 2 N: - 2 sin 30 = -1 N

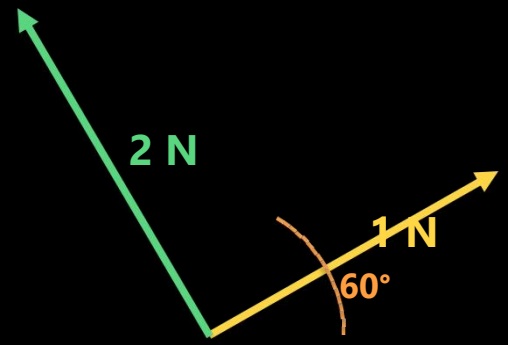
Step 3: Net x so far = -1/2 N. Add +1/2 N along +x to make $F_x = 0$

Option 4: 1/2 N.

Rough work / extra work

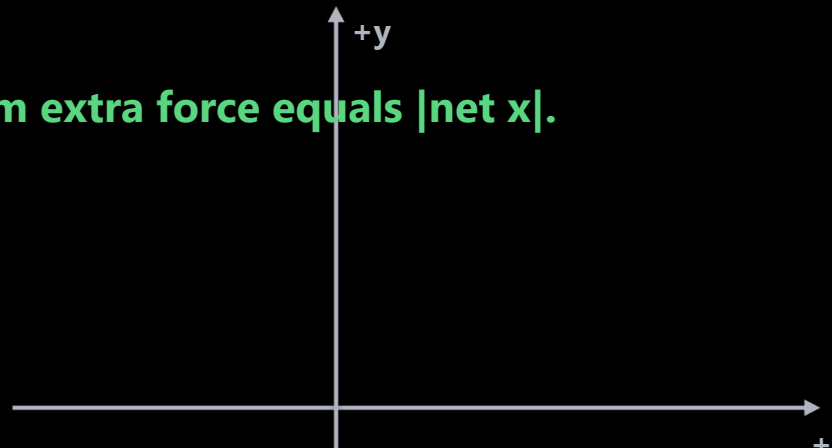
Cancel the leftover x-component

Keep y free. Kill only the net x by the smallest force.



Need F_x = 0

Extra force = 1/2 N along +x



Minimum extra force equals |net x|.

Direction of A + B from A

A = 6 î and B = 8 ĵ. Find A + B and its angle from A.

Solution: Direction of A + B from A

Step 1: $\vec{A} + \vec{B} = 6\,\hat{i} + 8\,\hat{j}$

Step 2: $\tan\theta = 8/6 = 4/3$, so $\theta = 53°$

Step 3: Dot check: $\cos\theta = (\vec{A} \cdot \vec{C}) / (A\,C) = 36 / (6 \times 10) = 3/5$

$\theta = 53°$ from A.

Three equal vectors at 60°

Place them in one plane with 60° between consecutive arrows.

Add any two first: R12 = 2 A cos(30°) = √3 A

That R12 is perpendicular to the third vector. Then |R| = 2A.

Do not treat all three as mutually at 120°. Consecutive angle is 60°.

Three vectors of magnitude A at 60°

Three vectors each of magnitude A meet at a point. Angle between consecutive vectors is 60°. |R| is:

1. 2A
2. √2 A
3. √3 A
4. √(6) A

Solution: Three vectors of magnitude A at 60°

Step 1: First pair: R12 = √3 A, and it is 90° from the third

Step 2: $|R| = \sqrt{(3\,A^2 + A^2)} = 2A$

Option 1: 2A.

Rough work / extra work

Force vectors from Coulomb's law

$F = k\, q_1\, q_2 / r^2$, along the line joining the charges

Same sign: repulsion. Opposite sign: attraction.

For equal magnitudes at 60°, R = √3 F. At 120°, R = F.

Draw arrows on the charge of interest first. Then add those arrows only.

Equilateral triangle, three +q

Three charges +q at the vertices of an equilateral triangle of side a. Find net force on any one charge.

Solution: Equilateral triangle, three +q

Step 1: Each pair force has magnitude $F = k q^2 / a^2$

Step 2: Angle between the two forces on one charge is 60°

Step 3: $F_{net} = 2 F \cos(30°) = \sqrt{3} F$

$F_{net} = \sqrt{3}\, k q^2 / a^2$.

Equilateral triangle with one -q

Charges +q, -q, +q at equilateral triangle side a. Find net force on the right-hand +q.

Solution: Equilateral triangle with one -q

Step 1: Both forces still have magnitude $F = k q^2 / a^2$

Step 2: One pulls left (attraction). One pushes out (repulsion). Angle between them is 120°

Step 3: For equal magnitudes at 120°, R = F

$F_{net} = k q^2 / a^2$.

Rough work / extra work

Net force on a corner charge of a square

Square side l. Charges: -q, +q, +q, and target +q. Find F_net on the target corner.

Solution: Net force on a corner charge of a square

Step 1: Side forces: $F_1 = F_2 = k\,q^2 / l^2$. Their resultant $F' = \sqrt{2}\,k\,q^2 / l^2$ outward

Step 2: Diagonal attraction: $F3 = k\,q^2 / (2\,l^2)$ inward along the diagonal

Step 3: $F_{net} = F' - F3 = (k\,q^2 / l^2)\,(\sqrt{2} - 1/2)$

$F_{net} = (k\,q^2 / l^2)\,[(2\,\sqrt{2} - 1)/2]$.

Net force on a block (ignore gravity)

Resolve to F_x and F_y. Then $R = \sqrt{(F_x^2 + F_y^2)}$.

$$\tan \alpha = |F_y| / |F_x|$$

α is measured from the x-axis toward F_net.

F_1 = 40 N, F_2 = 10 N, F3 = 30 N

On a block: F_1 = 40 N to +x, F_2 = 10 N to -x, F3 = 30 N to +y. Ignore gravity. Find F_net and its direction from x-axis.

Solution: F_1 = 40 N, F_2 = 10 N, F3 = 30 N

Step 1: $F_x = 40 - 10 = 30$ N, $F_y = 30$ N

Step 2: $F_{net} = \sqrt{(30^2 + 30^2)} = 30\sqrt{2}$ N

$\alpha = 45°$ above +x.

Rough work / extra work

F_1 = 60 N, F_2 = 20 N, F3 = 30 N

F_1 = 60 N (+x), F_2 = 20 N (-x), F3 = 30 N (+y). Ignore gravity and normal. Find F_net and α.

Solution: $F_1 = 60$ N, $F_2 = 20$ N, F3 = 30 N

Step 1: $F_x = 40$ N, $F_y = 30$ N

Step 2: F_net = $\sqrt{(40^2 + 30^2)} = 50$ N

Step 3: $\tan \alpha = 30/40 = 3/4$

$\alpha = 37°$ from +x.

Symmetry can force F_net = 0

Equal forces at equal spacing around a centre cancel in pairs.

Regular hexagon of equal masses: F_net on centre mass = 0

Three equal forces at 120° also cancel. Resultant of any two equals the third, opposite to it.

Equal distance is not enough if magnitudes are unequal.

Mass at centre of a regular hexagon

Equal masses m at the six vertices of a regular hexagon. Mass M is at the centre. Find gravitational F_net on M.

Solution: Mass at centre of a regular hexagon

Step 1: Opposite forces are equal and opposite

F_net on M = 0.

Rough work / extra work

Mass at centre of equilateral triangle

Equal masses m at the three vertices. Mass M at the centre. Find gravitational F_net on M.

Solution: Mass at centre of equilateral triangle

Step 1: Three equal forces, 120° apart

Step 2: Resultant of any two equals the third and cancels it

F_net on M = 0.

True or false: zero resultant

A. Resultant of two vectors lies in their plane. → True

B. Minimum number of unequal vectors for zero R is 2. → False

C. Minimum number of vectors for zero R is 2. → True (equal magnitude, opposite)

D. Minimum number of unequal vectors for zero R is 3. → True

E. Minimum number of non-coplanar vectors for zero R is 4. → True

Two unequal arrows cannot cancel. Three can close a triangle.

How scalars and vectors change

A scalar changes only when its magnitude changes.

A vector changes if magnitude changes, or direction changes, or both.

So Delta speed = 0 does not mean Delta v = 0.

Rough work / extra work

Change in velocity

$$\Delta \vec{v} = \vec{v}_f - \vec{v}_i$$

Always subtract vectors. Do not subtract speeds blindly.

$|\Delta \vec{v}|$ can be large even when $|\vec{v}|$ stays the same.

Change in speed = $|\vec{v}_f| - |\vec{v}_i|$. Change in velocity = $\vec{v}_f - \vec{v}_i$.

Rebound from a wall with same speed

A ball hits a wall with speed u along -î and rebounds with the same speed along +î. Find Delta v.

Solution: Rebound from a wall with same speed

Step 1: $\vec{v}_i = -u\,\hat{i}$, $\vec{v}_f = +u\,\hat{i}$

Step 2: $\Delta\vec{v} = u\,\hat{i} - (-u\,\hat{i}) = 2u\,\hat{i}$

$|\Delta\vec{v}| = 2u$. Change in speed = 0.

Hits the wall and stops

Same ball approaches with v_i = -u î and stops. Find Delta v.

Solution: Hits the wall and stops

Step 1: Delta v = 0 - (-u î) = u î

Delta v = u î.

Rough work / extra work

Deflected perpendicular, same speed

A particle with $v_i = u\,\hat{i}$ is deflected so that $v_f = u\,\hat{j}$. Find $|\Delta v|$ and Δp.

Solution: Deflected perpendicular, same speed

Step 1: Delta v = u ĵ - u î

Step 2: |Delta v| = √2 u

Delta p = m u ĵ - m u î.

After collision, moves along the incline

$v_i = u\,\hat{i}$. After collision, speed stays u and the path is along an incline of angle θ. Find Delta v.

Solution: After collision, moves along the incline

Step 1: $\vec{v}_f = u\cos\theta\,\hat{i} + u\sin\theta\,\hat{j}$

Step 2: $\Delta\vec{v} = u(\cos\theta - 1)\,\hat{i} + u\sin\theta\,\hat{j}$

Write components of the incline direction first.

Elastic bounce from a floor

Incident: u sin θ î - u cos θ ĵ. Rebound: u sin θ î + u cos θ ĵ. Find Delta v.

Solution: Elastic bounce from a floor

Step 1: Horizontal parts cancel

Step 2: Delta v = 2 u cos θ ĵ

Only the normal component reverses.

Rough work / extra work

Circular path, diametrically opposite

Speed is constant V on a circle. At a diametric opposite point, find change in speed and change in velocity.

Solution: Circular path, diametrically opposite

Step 1: Change in speed = 0

Step 2: If $v_i = V\hat{j}$ and $v_f = -V\hat{j}$, then $\Delta v = -2V\hat{j}$

$|\Delta v| = 2V$.

East then North, same speed

Initial velocity is 10 m/s east. Later it is 10 m/s north. Find Delta v.

Solution: East then North, same speed

Step 1: $\vec{v}_i = 10\,\hat{i}$, $\vec{v}_f = 10\,\hat{j}$

Step 2: $\Delta\vec{v} = -10\,\hat{i} + 10\,\hat{j}$

$|\Delta\vec{v}| = 10\sqrt{2}$ m/s.

Rough work / extra work

Relative velocity of two boys

Boy A moves at 6 m/s east. Boy B moves at 6 m/s at 30° east of north. Find velocity of B relative to A.

Solution: Relative velocity of two boys

Step 1: $\vec{v}_A = 6\,\hat{i}$

Step 2: $\vec{v}_B = 6\sin 30\,\hat{i} + 6\cos 30\,\hat{j} = 3\,\hat{i} + 3\sqrt{3}\,\hat{j}$

Step 3: $\vec{v}_{BA} = \vec{v}_B - \vec{v}_A = -3\,\hat{i} + 3\sqrt{3}\,\hat{j}$

$|\vec{v}_{BA}| = 6$ m/s.

North then east: three things to find

A particle goes from 5 m/s north to 5 m/s east. Find (i) Delta v (ii) |Delta v| (iii) change in magnitude of velocity.

Solution: North then east: three things to find

Step 1: Delta v = 5 î - 5 ĵ

Step 2: |Delta v| = 5 √2 m/s

Step 3: Change in magnitude of velocity = 5 - 5 = 0

Same speed, different direction: speed change is zero.

Position vector

$$\mathbf{r_{PQ} = r_P - r_Q}$$

This is the position of P with respect to Q.

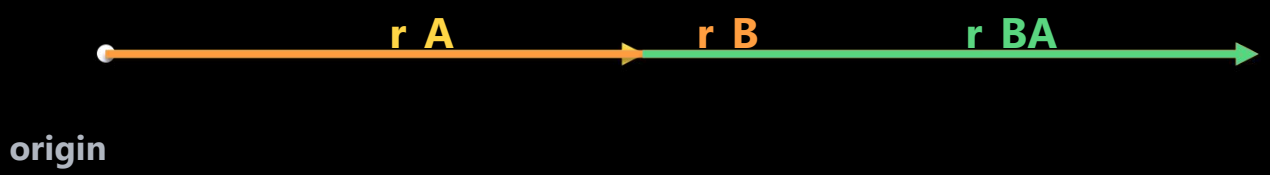
From origin: $\mathbf{r} = x\,\hat{\imath} + y\,\hat{\jmath} + z\,\hat{k}$.

Order matters. $\mathbf{r_{PQ}} = -\mathbf{r_{QP}}$.

Rough work / extra work

Relative position arrow

Start at the reference point. End at the object.



$$r_{BA} = r_B - r_A$$

Position from a reference point

An object is at (4, 3, 9). Find its position vector from the point (1, 7, 4).

Solution: Position from a reference point

Step 1: $\mathbf{r} = (4\,\hat{\imath} + 3\,\hat{\jmath} + 9\,\hat{k}) - (1\,\hat{\imath} + 7\,\hat{\jmath} + 4\,\hat{k})$

Step 2: $\mathbf{r} = 3\,\hat{\imath} - 4\,\hat{\jmath} + 5\,\hat{k}$

$|\mathbf{r}| = \sqrt{50} = 5\sqrt{2}$.

Find r_BA

A is at (2, 3, 0) and B is at (3, 0, 4). Find r_BA.

Solution: Find r_BA

Step 1: $\vec{r}_{BA} = \vec{r}_B - \vec{r}_A = (3\,\hat{i} + 4\,\hat{k}) - (2\,\hat{i} + 3\,\hat{j})$

$\vec{r}_{BA} = \hat{i} - 3\,\hat{j} + 4\,\hat{k}$.

Rough work / extra work

Find r_AB

B is at (1, 1, 1) and A is at (3, 0, 4). Find r_AB.

Solution: Find r_AB

Step 1: $\vec{r}_{AB} = \vec{r}_A - \vec{r}_B = (3\hat{i} + 4\hat{k}) - (\hat{i} + \hat{j} + \hat{k})$

$\vec{r}_{AB} = 2\hat{i} - \hat{j} + 3\hat{k}$.

Displacement versus distance

Displacement d = r_final - r_initial

If the object moves d1 then d2 then d3:

displacement = d1 + d2 + d3 , distance = |d1| + |d2| + |d3|

Distance is never negative. Displacement can cancel.

Position of A with respect to B

Find the position vector of A(1, 2, 1) with respect to B(-1, 2, 0).

Solution: Position of A with respect to B

Step 1: $\vec{r}_{AB} = \vec{r}_A - \vec{r}_B = (1\,\hat{i} + 2\,\hat{j} + \hat{k}) - (-\hat{i} + 2\,\hat{j})$

$\vec{r}_{AB} = 2\,\hat{i} + \hat{k}$.

Rough work / extra work

Three-step displacement (JEE Main style)

A student moves 50 m east, then 20 m north, then 40 √2 m south-east. Find net displacement.

Solution: Three-step displacement (JEE Main style)

Step 1: $\vec{d_1} = 50\,\hat{i}$, $\vec{d_2} = 20\,\hat{j}$

Step 2: $\vec{d_3} = 40\sqrt{2}(\cos 45^\circ\,\hat{i} - \sin 45^\circ\,\hat{j}) = 40\,\hat{i} - 40\,\hat{j}$

Step 3: $\vec{d} = 90\,\hat{i} - 20\,\hat{j}$

Net displacement $= 90\,\hat{i} - 20\,\hat{j}$.

Displacement from A to B in 3D

Initial position A(0, 4, 0). Final position B(3, 2, 4). Find displacement.

Solution: Displacement from A to B in 3D

Step 1: $\vec{d} = \vec{r}_f - \vec{r}_i = (3\,\hat{\imath} + 2\,\hat{\jmath} + 4\,\hat{k}) - 4\,\hat{\jmath}$

$$\vec{d} = 3\,\hat{\imath} - 2\,\hat{\jmath} + 4\,\hat{k}.$$

East, north, then south

A boy moves 30 m east, then 50 m north, then 10 m south. Find displacement and distance.

Solution: East, north, then south

Step 1: displacement = 30 $\hat{i}$ + 40 $\hat{j}$

Step 2: |displacement| = 50 m

Step 3: distance = 30 + 50 + 10 = 90 m

Displacement 50 m; distance 90 m.

Rough work / extra work

Dot product quick practice

$$A \cdot B = Ax\,Bx + Ay\,By + Az\,Bz$$

No diagram needed when components are given.

Result is a scalar. It can be negative.

Find the dot product

$A = 2\,\hat{i} + 3\,\hat{j} - 4\,\hat{k}$ and $B = 3\,\hat{i} - 2\,\hat{j} + 5\,\hat{k}$. Find $A \cdot B$.

Solution: Find the dot product

Step 1: $A \cdot B = (2)(3) + (3)(-2) + (-4)(5)$

Step 2: $= 6 - 6 - 20 = -20$

$A \cdot B = -20$.

Vector 05 lock lines

Vector = magnitude × chosen unit direction

Delta v = v_f - v_i ; change in speed is not the same thing

r_PQ = r_P - r_Q ; displacement = r_f - r_i

Distance adds path lengths. Displacement adds vectors.

Same speed does not mean zero Delta v.

Rough work / extra work

Practice

1. Redo the force-along-A problem without looking.

2. Redo three equal vectors at 60° and get |R| = 2A.

3. For east→north at equal speed, write Delta v and |Delta v|.

4. Separate displacement and distance for the 30 m, 50 m, 10 m walk.

Never subtract speeds when the question asks for change in velocity.